

Johansen Cointegration Test

Mathematical Notes & Journal

johansen.py — Dark Wing Project

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1 Why Do We Need the Johansen Test?

Consider two stock prices, say **Reliance** and **TCS**. Each of them individually wanders randomly over time — they are *non-stationary*. But what if there exists a magical *combination* of the two that does NOT wander? That combination is called a **cointegrating relationship**.

The **Johansen test** answers: “Does such a long-run equilibrium exist, and if so, how many independent equilibrium relationships are there?”

Advantage over Engle-Granger: Johansen handles $m \geq 2$ series simultaneously and can find multiple cointegrating vectors using a maximum likelihood framework.

2 Prerequisites: The I(1) Condition

2.1 Definition: Integrated of Order 1 — I(1)

A time series X_t is said to be **integrated of order 1**, written $X_t \sim I(1)$, if:

- X_t itself is *non-stationary* (it has a unit root, it wanders).
- Its *first difference* $\Delta X_t = X_t - X_{t-1}$ is *stationary* ($I(0)$).

Example: Log prices of stocks are typically $I(1)$. Daily returns (log differences) are $I(0)$.

Why it matters: The Johansen test is designed for $I(1)$ variables only. If you feed it $I(0)$ or $I(2)$ variables, the critical values are wrong and the results are meaningless.

This is verified in `johansen.py` using the `StationarityTest` class before running the test.

3 The VAR Foundation

3.1 The Vector Autoregression (VAR) Model

The Johansen test is derived from a p -th order **Vector Autoregression**:

$$X_t = \mu + A_1 X_{t-1} + A_2 X_{t-2} + \cdots + A_p X_{t-p} + \varepsilon_t \quad (1)$$

where:

- X_t is an $m \times 1$ vector of m time series at time t .
- μ is an $m \times 1$ vector of constants (intercepts).
- A_i are $m \times m$ coefficient matrices (how past values predict today's values).
- $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$ is a vector of white noise errors.
- p = the VAR lag order (selected by `VAROptimalLagSelect` in `Optimal_lag_selection.py`).

3.2 The VECM Reparameterisation

We can algebraically rewrite the VAR (1) into its **Vector Error Correction** form:

$$\Delta X_t = \mu + \Pi X_{t-1} + \Gamma_1 \Delta X_{t-1} + \Gamma_2 \Delta X_{t-2} + \cdots + \Gamma_{p-1} \Delta X_{t-p+1} + \varepsilon_t \quad (2)$$

where:

- $\Delta X_t = X_t - X_{t-1}$ are the first differences.
- $\Pi = \sum_{i=1}^p A_i - I_m$ is the **long-run impact matrix** (size $m \times m$).
- $\Gamma_j = -\sum_{i=j+1}^p A_i$ are short-run adjustment matrices.

The **entire Johansen test revolves around the matrix Π** . Specifically, its rank.

4 The Core Idea: Rank of Π

The rank of Π (denoted r) tells us exactly how many long-run equilibrium relationships exist:

Rank r	Meaning	Action
$r = 0$	No cointegration. $\Pi = 0$.	Use VAR in differences.
$0 < r < m$	r cointegrating vectors exist.	Fit VECM with rank r .
$r = m$	All series are stationary $I(0)$.	Use VAR in levels.

4.1 The Factorisation of Π

When $0 < r < m$, the matrix Π has a special factorisation:

$$\Pi = \alpha \beta^\top \quad (3)$$

- β is an $m \times r$ matrix. Each column is a **cointegrating vector** — the recipe for combining the m series into a stationary spread.
- α is an $m \times r$ matrix. Each column contains **adjustment speeds** — how fast each series corrects back to equilibrium.

Example for 2 series (Reliance & TCS):

$$\beta = \begin{pmatrix} 1 \\ -\beta_{12} \end{pmatrix}, \quad \text{so the spread is: } S_t = P_t^{REL} - \beta_{12} \cdot P_t^{TCS}$$

5 Johansen's MLE Estimation

5.1 Reduced Rank Regression

Johansen estimates $\Pi = \alpha \beta^\top$ using **Reduced Rank Regression (RRR)**, which is equivalent to solving a generalised eigenvalue problem.

Step 1 — Auxiliary Regressions:

Regress ΔX_t and X_{t-1} on the lagged differences $\{\Delta X_{t-1}, \dots, \Delta X_{t-p+1}\}$:

$$R_{0t} = \Delta X_t - \hat{B}_0 Z_t \quad (4)$$

$$R_{1t} = X_{t-1} - \hat{B}_1 Z_t \quad (5)$$

where $Z_t = (\Delta X_{t-1}^\top, \dots, \Delta X_{t-p+1}^\top)^\top$ is the vector of short-run regressors.

Step 2 — Moment Matrices:

$$S_{00} = \frac{1}{T} \sum_{t=1}^T R_{0t} R_{0t}^\top \quad (\text{variance of } \Delta X_t \text{ residuals}) \quad (6)$$

$$S_{11} = \frac{1}{T} \sum_{t=1}^T R_{1t} R_{1t}^\top \quad (\text{variance of } X_{t-1} \text{ residuals}) \quad (7)$$

$$S_{01} = \frac{1}{T} \sum_{t=1}^T R_{0t} R_{1t}^\top \quad (\text{cross-moment}) \quad (8)$$

Step 3 — Generalised Eigenvalue Problem:

Solve:

$$\boxed{|\lambda S_{11} - S_{10} S_{00}^{-1} S_{01}| = 0} \quad (9)$$

This yields m eigenvalues $\hat{\lambda}_1 > \hat{\lambda}_2 > \dots > \hat{\lambda}_m \geq 0$ and corresponding eigenvectors $\hat{v}_1, \hat{v}_2, \dots, \hat{v}_m$.

The **eigenvectors ARE the cointegrating vectors** β : the first r eigenvectors (those with largest eigenvalues) form the cointegrating space.

In code: `statsmodels.coint_johansen()` solves this exactly and returns the eigenvalues in `result.eig` and eigenvectors in `result.evec`.

6 The Two Test Statistics

6.1 Trace Statistic

Tests H_0 : rank $\leq r$ against H_1 : rank $> r$, for $r = 0, 1, \dots, m-1$:

$$\boxed{\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^m \ln(1 - \hat{\lambda}_i)} \quad (10)$$

Interpretation: If eigenvalues $\hat{\lambda}_{r+1}, \dots, \hat{\lambda}_m$ are all very small (close to 0), then $\ln(1 - \hat{\lambda}_i) \approx 0$, so the trace stat is small. We fail to reject H_0 and conclude rank $= r$.

Sequential procedure in code:

1. Test $r = 0$: Is $\lambda_{\text{trace}}(0) > \text{critical value}$? If Yes, reject, move to $r = 1$.
2. Test $r = 1$: Is $\lambda_{\text{trace}}(1) > \text{critical value}$? If No, STOP. Rank $= 1$.
3. Continue until the first failure to reject. That r is the rank.

6.2 Maximum Eigenvalue Statistic

Tests H_0 : rank = r against H_1 : rank = $r + 1$:

$$\lambda_{\max}(r, r + 1) = -T \ln(1 - \hat{\lambda}_{r+1}) \quad (11)$$

This tests *one additional* cointegrating vector at a time. It is more specific than the trace test. Both tests should agree. If they disagree, the trace test is generally considered more robust.

6.3 Critical Values

Critical values depend on:

- $m - r$ (the number of series tested under the null).
- The deterministic specification (`det_order`).
- The chosen significance level (1%, 5%, or 10%).

Critical values are non-standard (cannot use the normal χ^2 table) because the test statistics involve Brownian motion functionals. Osterwald-Lenum (1992) tables are used. `statsmodels` provides these in `result.cvt` (trace) and `result.cvm` (max-eig).

7 Deterministic Terms (`det_order`)

<code>det_order</code>	Meaning	When to use
-1	No intercept, no trend	Prices centred at 0 (rare)
0	Restricted constant (intercept in CE only)	Log prices (most common)
1	Restricted linear trend	Series with drift (GDP, interest rates)

For stock log prices, use `det_order = 0` (restricted constant), which is the default in `johansen.py`.

8 The Lag Relationship: p vs. k_ar_diff

The Johansen test is parameterised by k_ar_diff , the number of lagged differences in the VECM. The relationship to the VAR lag p is:

$$k_ar_diff = p - 1 \quad (12)$$

Example: If `VAROptimalLagSelect` returns $p^* = 3$, then $k_ar_diff = 2$.

This is automated in `johansen.py`: when `k_ar_diff=None`, it calls `VAROptimalLagSelect` internally.

9 How to Interpret the Output: Worked Example

Suppose we test 3 series (Reliance, TCS, Infosys) at 5% significance.

Trace test output:

H_0	Trace Stat	CV (5%)	Reject?
$r \leq 0$	48.32	29.80	Yes (accept)
$r \leq 1$	14.11	15.49	No
$r \leq 2$	3.20	3.84	No

Reading:

- $r = 0$: Rejected. At least 1 cointegrating vector exists.
- $r = 1$: Failed to reject. At most 1 cointegrating vector.
- **Conclusion: Cointegrating rank $r^* = 1$.**

The single cointegrating vector β (extracted from the eigenvector with the largest eigenvalue) gives the recipe for the stationary spread:

$$S_t = \beta_1 \cdot \text{REL}_t + \beta_2 \cdot \text{TCS}_t + \beta_3 \cdot \text{INFY}_t$$

This spread S_t is $I(0)$ (stationary) and is what you monitor for mean-reversion trading.

10 Connection to the Dark Wing Pipeline

Step 1: Unit Root: Confirm all series are $I(1)$ via `StationarityTest.check_stationarity()`.

Step 2: Lag Selection: Get p^* from `VAROptimalLagSelect.select_lag()`.

Step 3: Johansen: Run `JohansenTest.run()`, obtain rank r^* and β .

Step 4: VECM: Fit `fit_vecm.py` using rank r^* , extract α (adjustment speeds).

Step 5: Kalman: Use β as the initial hedge ratio in `dynamic_beta.py`.

Step 6: Signal: Compute spread $S_t = \beta^\top X_t$, z-score it, generate buy/sell signal.

11 Mathematical Derivation: Why the Eigenvalue Problem?

This section shows, step by step, why solving $|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}| = 0$ finds the cointegrating vectors. You can skip this in practice but it is good to know.

We want to maximise the log-likelihood of the VECM subject to the rank constraint $\text{rank}(\Pi) = r$.

The concentrated log-likelihood (after profiling out α and Γ_j) is:

$$\mathcal{L}(\beta) = \text{const} - \frac{T}{2} \ln |S_{00} - S_{01}\beta(\beta^\top S_{11}\beta)^{-1}\beta^\top S_{10}| \quad (13)$$

Maximising this over β subject to the normalisation $\beta^\top S_{11}\beta = I_r$ leads to the eigenvalue equation (9).

The r eigenvectors with the **largest** eigenvalues $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_r$ are the Maximum Likelihood Estimator of β .

Each eigenvalue $\hat{\lambda}_i$ measures the *strength* of the i -th cointegrating relationship — how tightly the corresponding linear combination is pulled back to its mean.